

To configure MP-BGP EVPN on each VTEP:**1. Configure the EVPN settings:**

```

config system evpn
  edit <id>
    set rd {AA | AA:NN | A.B.C.D:NN}
    set import-rt <AA:NN>
    set export-rt <AA:NN>
    set ip-local-learning {enable | disable}
    set arp-suppression {enable | disable}
  next
end

```

The `ip-local-learning` setting is used to enable/disable monitoring the local ARP table of the switch interface to learn the IP/MAC bindings, and advertise them to neighbors. This setting is disabled by default, but must be enabled when configuring MP-BGP EVPN.

The `arp-suppression` setting is used to enable/disable using proxy ARP to perform suppression of ARP discovery using the flood-and-learn approach. This setting is disabled by default. When enabled, proxy ARP entries are added on the switch interface to suppress the ARP flooding of known IP/MAC bindings, which were learned by the MP-BGP EVPN control plane.

2. Configure the EVPN settings within the VXLAN settings:

```

config system vxlan
  edit <name>
    set interface <string>
    set vni <integer>
    set evpn-id <integer>
    set learn-from-traffic {enable | disable}
  next
end

```

The `learn-from-traffic` setting is used to enable/disable learning of remote VNIs from VXLAN traffic. This setting is disabled by default, and should only be enabled when local and all remote peers are using same VNI value, and some of the peers do not have MP-BGP EVPN capability.

3. Configure the BGP settings:

```

config router bgp
  set ibgp-multipath {enable | disable}
  set recursive-next-hop {enable | disable}
  set graceful-restart {enable | disable}
  config neighbor
    edit <WAN_IP_of_other_VTEP>
      set ebgp-enforce-multi-hop {enable | disable}
      set next-hop-self {enable | disable}
      set next-hop-self-vpnv4 {enable | disable}
      set soft-reconfiguration {enable | disable}
      set soft-reconfiguration-evpn {enable | disable}
      set remote-as <AS_number>
    next
  end
end

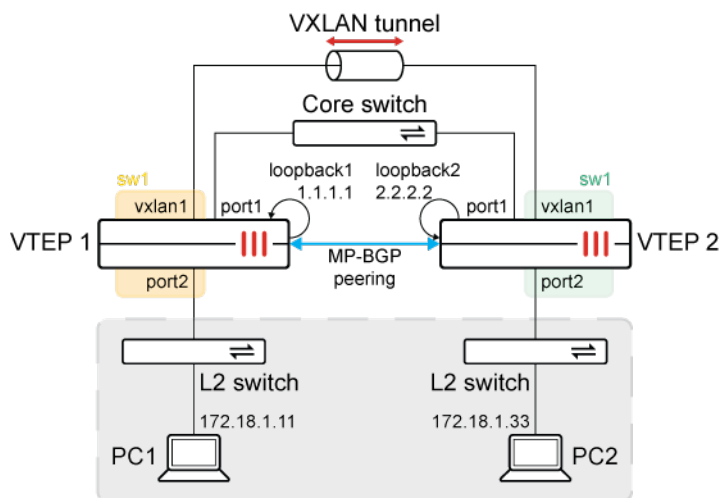
```

4. Configure the EVPN setting within the HA settings:

```
config system ha
    set evpn-ttl <integer>
end
```

Example

In this example, two FortiGates are configured as VXLAN tunnel endpoints (VTEPs). A VXLAN is configured to allow L2 connectivity between the networks behind each FortiGate. The VXLAN interface vxlan1 and port2 are placed on the same L2 network using a software switch (sw1). An L2 network is formed between PC1 and PC2. MP-BGP EVPN is used as the control plane to learn and distribute MAC address information within a single L2 domain identified using a specific VNI.



The VTEPs have the following MAC address tables:

Interface/endpoint	VTEP1	VTEP2
vxlan1	82:51:d1:44:bf:93	d2:21:00:c9:e6:98
port2	50:00:00:03:00:01	50:00:00:04:00:01
sw1	50:00:00:03:00:01	50:00:00:04:00:01

The MAC address of PC1 is 00:50:00:00:06:00. The MAC address of PC2 is 00:50:00:00:07:00.

This example assumes that the WAN interface and default route settings have already been configured on the VTEP 1 and VTEP 2 FortiGates. These configurations are omitted from the example. All peers are configured for MP-BGP EVPN.

To configure the VTEP1 FortiGate:

1. Configure the loopback interface:

```
config system interface
    edit "loopback1"
        set vdom "root"
    end
```

```
        set ip 1.1.1.1 255.255.255.255
        set allowaccess ping https ssh http
        set type loopback
    next
end
```

2. Configure the EVPN settings:

```
config system evpn
    edit 100
        set rd "100:100"
        set import-rt "1:1"
        set export-rt "1:1"
        set ip-local-learning enable
        set arp-suppression enable
    next
end
```

3. Configure the local interface and EVPN settings within the VXLAN settings:

```
config system vxlan
    edit "vxlan1"
        set interface "loopback1"
        set vni 1000
        set evpn-id 100
    next
end
```

4. Configure the EVPN settings within the BGP settings:

```
config router bgp
    set as 65001
    set router-id 1.1.1.1
    set ibgp-multipath enable
    set recursive-next-hop enable
    set graceful-restart enable
    config neighbor
        edit "172.25.160.101"
            set ebgp-enforce-multi-hop enable
            set next-hop-self enable
            set next-hop-self-vpnv4 enable
            set soft-reconfiguration enable
            set soft-reconfiguration-evpn enable
            set remote-as 65001
        next
    end
    config network
        edit 1
            set prefix 1.1.1.1 255.255.255.255
        next
    end
end
```

172.27.16.237 is the WAN IP address of the VTEP2 FortiGate.

5. Configure the software switch:

```
config system switch-interface
  edit "sw1"
    set vdom "root"
    set member "port2" "vxlan1"
    set intra-switch-policy explicit
  next
end
```

6. Configure the software switch interface settings:

```
config system interface
  edit "sw1"
    set vdom "root"
    set ip 172.18.1.253 255.255.255.0
    set allowaccess ping
    set type switch
  next
end
```

7. Configure the firewall policies between the member interfaces in the software switch:

```
config firewall policy
  edit 1
    set srcintf "port2"
    set dstintf "vxlan1"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
  next
  edit 2
    set srcintf "vxlan1"
    set dstintf "port2"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
  next
end
```

To configure the VTEP2 FortiGate:

1. Configure the loopback interface:

```
config system interface
  edit "loopback2"
    set vdom "root"
```

```
        set ip 2.2.2.2 255.255.255.255
        set allowaccess ping https ssh http
        set type loopback
    next
end
```

2. Configure the EVPN settings:

```
config system evpn
    edit 100
        set rd "100:100"
        set import-rt "1:1"
        set export-rt "1:1"
        set ip-local-learning enable
        set arp-suppression enable
    next
end
```

3. Configure the local interface and EVPN settings within the VXLAN settings:

```
config system vxlan
    edit "vxlan1"
        set interface "loopback2"
        set vni 1000
        set evpn-id 100
    next
end
```

4. Configure the EVPN settings within the BGP settings:

```
config router bgp
    set as 65001
    set router-id 2.2.2.2
    set ibgp-multipath enable
    set recursive-next-hop enable
    set graceful-restart enable
    config neighbor
        edit "172.25.160.100"
            set ebgp-enforce-multihop enable
            set next-hop-self enable
            set next-hop-self-vpnv4 enable
            set soft-reconfiguration enable
            set soft-reconfiguration-evpn enable
            set remote-as 65001
        next
    end
    config network
        edit 1
            set prefix 2.2.2.2 255.255.255.255
        next
    end
end
```

172.27.16.236 is the WAN IP address of the VTEP1 FortiGate.

5. Configure the software switch:

```
config system switch-interface
  edit "sw1"
    set vdom "root"
    set member "port2" "vxlan1"
    set intra-switch-policy explicit
  next
end
```

6. Configure the software switch interface settings:

```
config system interface
  edit "sw1"
    set vdom "root"
    set ip 172.18.1.254 255.255.255.0
    set allowaccess ping
    set type switch
  next
end
```

7. Configure the firewall policies between the member interfaces in the software switch:

```
config firewall policy
  edit 1
    set srcintf "port2"
    set dstintf "vxlan1"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
  next
  edit 2
    set srcintf "vxlan1"
    set dstintf "port2"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
  next
end
```

To verify the MP-BGP EVPN status on the VTEP1 FortiGate:

1. From a host computer with IP address 172.18.1.11, perform the following.
 - a. Check the ARP cache:

```
# arp
Address                Hwtype  Hwaddress      Flags Mask      Iface
172.18.1.253           ether   50:00:00:03:00:01  C              ens3
```

- b. Ping the host computer with IP address 172.18.1.33:

```
# ping 172.18.1.33 -c 4
PING 172.18.1.33 (172.18.1.33) 56(84) bytes of data.
64 bytes from 172.18.1.33: icmp_seq=1 ttl=64 time=1325 ms
64 bytes from 172.18.1.33: icmp_seq=2 ttl=64 time=319 ms
64 bytes from 172.18.1.33: icmp_seq=3 ttl=64 time=3.96 ms
64 bytes from 172.18.1.33: icmp_seq=4 ttl=64 time=1.66 ms

--- 172.18.1.33 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 1.660/412.614/1325.209/542.530 ms
```

- c. Check the ARP cache again:

```
# arp
Address                Hwtype  Hwaddress      Flags Mask      Iface
172.18.1.33            ether   00:50:00:00:07:00  C              ens3
172.18.1.253           ether   50:00:00:03:00:01  C              ens3
```

2. On the VTEP1 FortiGate, run the switch and VXLAN debug commands.

- a. Verify the forwarding database for vxlan1:

```
# diagnose sys vxlan fdb list vxlan1
mac=00:00:00:00:00:00 state=0x0082 remote_ip=2.2.2.2 port=4789 vni=1000 ifindex0
mac=00:50:00:00:07:00 state=0x0082 remote_ip=2.2.2.2 port=4789 vni=1000 ifindex0

total fdb num: 2
```

- b. Verify the forwarding database statistics for vxlan1:

```
# diagnose sys vxlan fdb stat vxlan1
fdb_table_size=256 fdb_table_used=2 fdb_entry=2 fdb_max_depth=1 cleanup_idx=0 c2
```

- c. Verify the bridging information for sw1:

```
# diagnose netlink brctl name host sw1
show bridge control interface sw1 host.
fdb: hash size=32768, used=5, num=5, depth=1, gc_time=4, ageing_time=3, arp-sup
Bridge sw1 host table
port no device devname mac addr          ttl    attributes
2    15    vxlan1  00:00:00:00:00:00      28     Hit(28)
2    15    vxlan1  00:50:00:00:07:00      18     Hit(18)
2    15    vxlan1  82:51:d1:44:bf:93      0      Local Static
1    4     port2  00:50:00:00:06:00      14     Hit(14)
1    4     port2  50:00:00:03:00:01      0      Local Static
```

3. Run the BGP EVPN commands and observe the route type 2 (MAC/IP advertisement route) and route type 3 (inclusive multicast Ethernet tag route).

a. Verify the BGP L2 VPN EVPN summary information:

```
# get router info bgp evpn summary

VRF 0 BGP router identifier 1.1.1.1, local AS number 65001
BGP table version is 2
1 BGP AS-PATH entries
0 BGP community entries

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/Pd
172.25.160.101 4      65001      9      9        1    0    0 00:04:02      3

Total number of neighbors 1
```

b. Verify the BGP L2 VPN EVPN network information:

```
# get router info bgp evpn network

Network      Next Hop      Metric      LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (Default for VRF 0)
*> [2][0][48][00:50:00:00:06:00][0]/72
      1.1.1.1      0      100 32768      0 i <-/>
*> [2][0][48][00:50:00:00:06:00][32][172.18.1.11]/104
      1.1.1.1      0      100 32768      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][0]/72
      2.2.2.2      0      100      0      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][32][172.18.1.33]/104
      2.2.2.2      0      100      0      0 i <-/>
*> [3][0][32][1.1.1.1]/80
      1.1.1.1      0      100 32768      0 i <-/>
*>i[3][0][32][2.2.2.2]/80
      2.2.2.2      0      100      0      0 i <-/>

Network      Next Hop      Metric      LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (received from VRF 0)
*>i[2][0][48][00:50:00:00:07:00][0]/72
      2.2.2.2      0      100      0      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][32][172.18.1.33]/104
      2.2.2.2      0      100      0      0 i <-/>
*>i[3][0][32][2.2.2.2]/80
      2.2.2.2      0      100      0      0 i <-/>
```

c. Verify the BGP L2 VPN EVPN context:

```
# get router info bgp evpn context
L2VPN EVPN context for VRF 0
ID 100 vlan-based, RD is [100:100]
Import RT: RT:1:1
Export RT: RT:1:1
Bridge domain 0 VNI 1000
Encapsulation 8(VXLAN)
Source interface loopback1
Source address 1.1.1.1
```

d. Verify the BGP L2 VPN EVPN information for VRF 0:


```
# get router info bgp evpn vrf 0
  Network          Next Hop          Metric      LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (Default for VRF 0)
*> [2][0][48][00:50:00:00:06:00][0]/72
      1.1.1.1          0          100  32768      0 i <-/>
*> [2][0][48][00:50:00:00:06:00][32][172.18.1.11]/104
      1.1.1.1          0          100  32768      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][0]/72
      2.2.2.2          0          100    0      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][32][172.18.1.33]/104
      2.2.2.2          0          100    0      0 i <-/>
*> [3][0][32][1.1.1.1]/80
      1.1.1.1          0          100  32768      0 i <-/>
*>i[3][0][32][2.2.2.2]/80
      2.2.2.2          0          100    0      0 i <-/>

  Network          Next Hop          Metric      LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (received from VRF 0)
*>i[2][0][48][00:50:00:00:07:00][0]/72
      2.2.2.2          0          100    0      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][32][172.18.1.33]/104
      2.2.2.2          0          100    0      0 i <-/>
*>i[3][0][32][2.2.2.2]/80
      2.2.2.2          0          100    0      0 i <-/>
```

- e. Verify the BGP L2 VPN EVPN information for RD 100:100:

```
# get router info bgp evpn rd 100:100
  Network          Next Hop          Metric      LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (Default for VRF 0)
*> [2][0][48][00:50:00:00:06:00][0]/72
      1.1.1.1          0          100  32768      0 i <-/>
*> [2][0][48][00:50:00:00:06:00][32][172.18.1.11]/104
      1.1.1.1          0          100  32768      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][0]/72
      2.2.2.2          0          100    0      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][32][172.18.1.33]/104
      2.2.2.2          0          100    0      0 i <-/>
*> [3][0][32][1.1.1.1]/80
      1.1.1.1          0          100  32768      0 i <-/>
*>i[3][0][32][2.2.2.2]/80
      2.2.2.2          0          100    0      0 i <-/>

  Network          Next Hop          Metric      LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (received from VRF 0)
*>i[2][0][48][00:50:00:00:07:00][0]/72
      2.2.2.2          0          100    0      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][32][172.18.1.33]/104
      2.2.2.2          0          100    0      0 i <-/>
*>i[3][0][32][2.2.2.2]/80
      2.2.2.2          0          100    0      0 i <-/>
```

- f. Verify the neighbor EVPN advertised routes for 172.25.160.101:

```
# get router info bgp neighbors 172.25.160.101 advertised-routes evpn
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (Default for VRF 0) (Default for VRF 0)
*>i[2][0][48][00:50:00:00:06:00][0]/72
                1.1.1.1                100  32768      0 i <-/>
*>i[2][0][48][00:50:00:00:06:00][32][172.18.1.11]/104
                1.1.1.1                100  32768      0 i <-/>
*>i[3][0][32][1.1.1.1]/80
                1.1.1.1                100  32768      0 i <-/>

Total number of prefixes 3
```

- g. Verify the neighbor EVPN received routes for 172.25.160.101:

```
# get router info bgp neighbors 172.25.160.101 received-routes evpn
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (received from VRF 0) (received from VRF 0)
*>i[2][0][48][00:50:00:00:07:00][0]/72
                2.2.2.2                100    0      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][32][172.18.1.33]/104
                2.2.2.2                100    0      0 i <-/>
*>i[3][0][32][2.2.2.2]/80
                2.2.2.2                100    0      0 i <-/>

Total number of prefixes 3
```

- h. Verify the neighbor EVPN routes:

```
# get router info bgp neighbors 172.25.160.101 routes evpn
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (Default for VRF 0) (Default for VRF 0)
*>i[2][0][48][00:50:00:00:07:00][0]/72
                2.2.2.2                  0      100    0      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][32][172.18.1.33]/104
                2.2.2.2                  0      100    0      0 i <-/>
*>i[3][0][32][2.2.2.2]/80
                2.2.2.2                  0      100    0      0 i <-/>
Route Distinguisher: 100:100 (received from VRF 0) (received from VRF 0)
*>i[2][0][48][00:50:00:00:07:00][0]/72
                2.2.2.2                  0      100    0      0 i <-/>
*>i[2][0][48][00:50:00:00:07:00][32][172.18.1.33]/104
```

```

                2.2.2.2          0          100      0          0 i <-/>
*>i[3][0][32][2.2.2.2]/80
                2.2.2.2          0          100      0          0 i <-/>

Total number of prefixes 6

```

4. Run the following EVPN get commands.

a. Verify the EVPN instances:

```

# get l2vpn evpn instance
EVPN instance: 100
IP local learning enabled
ARP suppression enabled
HA primary
  Number of bridge domain: 1
  Bridge domain: TAGID 0 VNI 1000 ADDR 1.1.1.1 VXLAN vxlan1 SWITCH sw1

```

b. Verify the EVPN table:

```

# get l2vpn evpn table
EVPN instance 100
Broadcast domain VNI 1000 TAGID 0

  EVPN instance 100
  Broadcast domain VNI 1000 TAGID 0

  EVPN MAC table:
  MAC          VNI      Remote Addr    Binded Address
  00:50:00:00:07:00 1000    2.2.2.2      172.18.1.33
                  1000    2.2.2.2      -

  EVPN IP table:
  Address      VNI      Remote Addr    MAC
  172.18.1.33  1000    2.2.2.2      00:50:00:00:07:00

  EVPN Local MAC table:
  "Inactive" means this MAC/IP pair will not be sent to peer.
  Flag code: S - Static F - FDB. Trailing * means HA
  MAC          Flag Status  Binded Address
  00:50:00:00:06:00  Active  172.18.1.11
                  F    Active  -

  EVPN Local IP table:
  Address      MAC
  172.18.1.11  00:50:00:00:06:00

  EVPN PEER table:
  VNI      Remote Addr    Binded Address
  1000     2.2.2.2      2.2.2.2

```

5. Run the proxy ARP diagnose command:

```
# diagnose ip arp list
Address      Hardware Addr  Interface
172.18.1.33   00:50:00:00:07:00 sw1
```

EVPN support for VXLAN Anycast gateway and type-5 routing **NEW**

This feature adds support for VXLAN Anycast gateway, EVPN IRB (Ethernet VPN Integrated Routing and Bridging), and type-5 EVPN routes, enabling seamless L2/L3 service integration and scalable IP prefix advertisement across the VXLAN fabric. It simplifies mobility and routing in modern data center deployments. For more information about EVPN, see [VXLAN with MP-BGP EVPN on page 332](#).

The `config router bgp` command has the following relevant options:

```
config router bgp
  config redistribute {connected | static | isis | ospf | rip}
    set status-evpn {enable | disable}
    set route-map-evpn <string>
  end
  config {neighbor | neighbor-group}
    edit <string>
      set adv-evpn-route {type2 | type5 | local}
    next
  end
end
```

Option	Description
<code>config redistribute {connected static isis ospf rip}</code>	Types of routes to be redistributed.
<code>status-evpn {enable disable}</code>	Enable/disable EVPN redistribution status.
<code>route-map-evpn <string></code>	Route map name for EVPN redistribution.
<code>adv-evpn-route {type2 type5 local}</code>	Types of EVPN routes that can be advertised to this neighbor as IPv4 routes: <code>type2</code>: Type 2 EVPN routes. <code>type5</code>: Type 5 EVPN routes. <code>local</code>: Local EVPN routes.

The `config system evpn` command has the following relevant options:

```
config system evpn
  edit <id>
    set type {macvrf | ipvrf}
    set adv-default-gw {enable | disable}
    set virtual-mac-vrid <integer>
    set l3-instance <integer>
```

```
set *interface <string>
set distribute-local-route {enable | disable}
next
end
```

Option	Description
type {macvrf ipvrf}	EVPN instance type: • macvrf: MAC VRF. • ipvrf: L3 VRF.
adv-default-gw {enable disable}	Enable/disable advertisement of default gateway.
virtual-mac-vrid <integer>	VRID used to generate virtual MAC address (1 to 255).
l3-instance <integer>	L3VRF EVPN instance ID (0 to 4294967295).
*interface <string>	Outgoing interface for all VxLANs in this L3VRF EVPN instance. Available when type is set to ipvrf.
distribute-local-route {enable disable}	Enable/disable distribution of local EVPN routes as IPv4 routes to BGP.

A new get command is available for default-gateway:

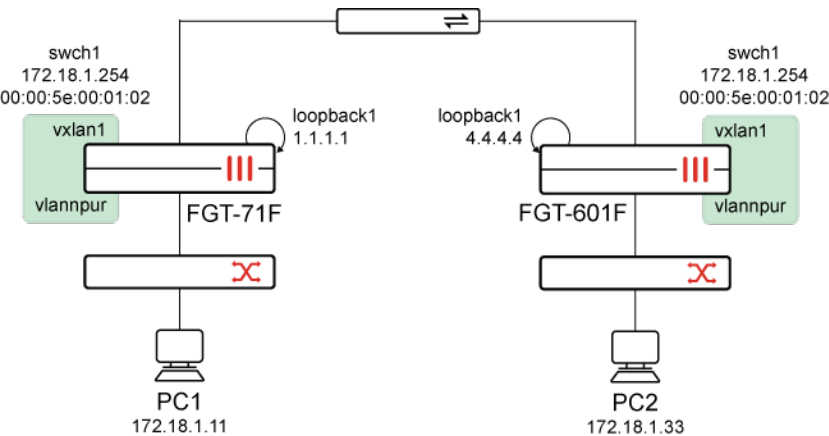
```
get l2vpn evpn default-gateway
```

Example

This section includes the following examples:

- [Anycast gateway example on page 346](#)
- [IRB example on page 351](#)
- [Type-5 route example on page 356](#)

Anycast gateway example



In a Security Fabric configured with EVPN, VXLAN, and an Anycast gateway (swch1), all leaf switches that connect endpoints in the same VLAN (or subnet) share the same virtual default gateway IP address. The leaf switches also share the same virtual MAC address for that gateway.

Endpoints (such as servers, VMs, containers) always see the same gateway regardless of which leaf they connect to. If a VM or host moves from Leaf1 to Leaf2, its default gateway stays the same and doesn't require any reconfiguration (for example re-ARP or a new IP address), which provides seamless mobility.

This example describes how to configure FortiGate-71F and FortiGate-601F in the topology. It also describes how to verify the configuration.

To configure FortiGate-71F:

1. Configure EVPN:

```
config system evpn
  edit 100
    set rd "100:100"
    set import-rt "1:1"
    set export-rt "1:1"
    set ip-local-learning enable
    set adv-default-gw enable
    set virtual-mac-vrid 2
  next
end
```

2. Configure VXLAN:

```
config system vxlan
  edit "vxlan1"
    set interface "loopback1"
    set vni 1000
    set evpn-id 100
    set learn-from-traffic enable
  next
end
```

3. Configure a system switch interface:

```
config system switch-interface
  edit "swch1"
    set vdom "root"
    set member "vxlan1" "vlannpur"
  next
end
```

4. Configure a system interface:

```
config system interface
  edit "swch1"
    set vdom "root"
    set ip 172.18.1.254 255.255.255.0
    set allowaccess ping
```

```
        set type switch
        set snmp-index 26
    next
end
```

5. Configure BGP:

```
config router bgp
    set as 65412
    set router-id 1.1.1.1
    set ibgp-multipath enable
    set network-import-check disable
    set recursive-next-hop enable
    set graceful-restart enable
    config neighbor
        edit "172.18.5.123"
            set ebgp-enforce-multihop enable
            set next-hop-self enable
            set next-hop-self-vpnv4 enable
            set soft-reconfiguration enable
            set soft-reconfiguration-evpn enable
            set remote-as 65412
            set graceful-shutdown-community "22:55"
        next
    end
    config network
        edit 1
            set prefix 1.1.1.1 255.255.255.255
        next
        edit 4
            set prefix 172.28.6.0 255.255.255.0
        next
    end
    config redistribute "connected"
    end
    config redistribute "rip"
    end
    config redistribute "ospf"
    end
    config redistribute "static"
    end
    config redistribute "isis"
    end
    config redistribute6 "connected"
    end
    config redistribute6 "rip"
    end
    config redistribute6 "ospf"
    end
    config redistribute6 "static"
    end
    config redistribute6 "isis"
```

```
    end
end
```

To configure FortiGate-601F:**1. Configure EVPN:**

```
config system evpn
  edit 100
    set rd "100:100"
    set import-rt "1:1"
    set export-rt "1:1"
    set ip-local-learning enable
    set adv-default-gw enable
    set virtual-mac-vrid 2
  next
end
```

2. Configure VXLAN:

```
config system vxlan
  edit "vxlan1"
    set interface "loopback1"
    set vni 1000
    set evpn-id 100
    set learn-from-traffic enable
  next
end
```

3. Configure a system switch interface:

```
config system switch-interface
  edit "swch1"
    set vdom "root"
    set member "vxlan1" "vlannpur"
  next
end
```

4. Configure a system interface:

```
config system interface
  edit "swch1"
    set vdom "root"
    set ip 172.18.1.254 255.255.255.0
    set allowaccess ping
    set type switch
    set snmp-index 50
  next
end
```

5. Configure BGP:


```
config router bgp
  set as 65412
  set router-id 4.4.4.4
  set ibgp-multipath enable
  set recursive-next-hop enable
  config neighbor
    edit "172.18.5.121"
      set next-hop-self enable
      set next-hop-self-vpnv4 enable
      set soft-reconfiguration enable
      set soft-reconfiguration-evpn enable
      set remote-as 65412
    next
  end
  config network
    edit 1
      set prefix 4.4.4.4 255.255.255.255
    next
  end
  config redistribute "connected"
  end
  config redistribute "rip"
  end
  config redistribute "ospf"
  end
  config redistribute "static"
  end
  config redistribute "isis"
  end
  config redistribute6 "connected"
  end
  config redistribute6 "rip"
  end
  config redistribute6 "ospf"
  end
  config redistribute6 "static"
  end
end
```

To verify:

1. From FortiGate-601F, ping 172.18.1.11:

```
# execute ping 172.18.1.11
PING 172.18.1.11 (172.18.1.11): 56 data bytes
64 bytes from 172.18.1.11: icmp_seq=0 ttl=255 time=0.4 ms
64 bytes from 172.18.1.11: icmp_seq=1 ttl=255 time=0.3 ms
^C
--- 172.18.1.11 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.3/0.3/0.4 ms

FortiGate-601F (vdom1) (Interim)# diagnose ip arp list | grep 254
```

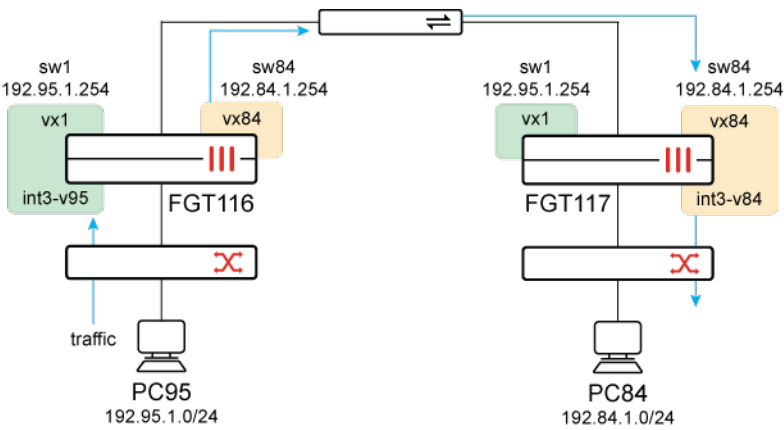
```
index=51 ifname=vlanpu1 172.18.1.254 00:00:5e:00:01:02 state=00000004 use=9094399
confirm=9100246 update=9094246 ref=0

FortiGate-601F (vdom1) (Interim)#
```

2. From FortiGate-71F, diagnose the IP ARP list:

```
# diagnose ip arp list
index=27 ifname=vlanpu1 172.18.1.33 86:39:8f:13:f7:01 state=00000002 use=2374 confirm=2374
update=1946 ref=3
index=27 ifname=vlanpu1 172.18.1.254 00:00:5e:00:01:02 state=00000002 use=435 confirm=435
update=435 ref=2
version count: 2
```

IRB example



IRB (Integrated Routing and Bridging) in EVPN means that FortiGate can both bridge (L2) and route (L3) traffic inside the Security Fabric configured with VXLAN/EVPN. EVPN IRB allows L2 services for intra-subnet communication and L3 services for inter-subnet communications.

This example uses the following VLAN, subnets, and VNI names:

PC	VLAN	Subnet	VNI
PC95	vlan95	192.95.1.0/24	vni 141
PC94	vlan84	192.84.1.0/24	vni 84

This example describes how to configure FGT116 and FGT117 in the topology to allow inter-subnet routing between vlan95 and vlan84 over the VXLANs between the two FortiGate VTEPs. This allows PC95 to reach PC84 using EVPN over VXLAN.

To configure FGT116:

1. Configure EVPN:

```
config system evpn
edit 1
```